From MAP Preservation to Posterior-Moment Fidelity in Coded SP-AFDM

V2 handoff audit, two method-synthesis tests, and a conditional list-refinement theory

Research continuation and reproducible numerical study

19 September 2026

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1. Research verdict

Selected research line: posterior-fidelity-controlled, code-aided channel estimation. The concrete reference construction is a bounded codeword list, full-channel likelihood rescoring, projected posterior moments, and a conditional channel-refit error guard. This replaces the previous emphasis on preserving only the terminal MAP decision when pruning a Gram matrix.

The decisive distinction is that a distribution adequate for choosing a codeword need not be adequate for supplying channel-estimation moments. A correct MAP word in a singleton list gives zero conditional covariance, even when the true posterior retains considerable mass elsewhere. Conversely, including more words can correct the moments without changing a single hard decision. This is relevant to the project's actual mean/variance feedback interface, rather than an additional detector block without an identified consumer.

Publication assessment: no new receiver has yet satisfied novelty, complete computational cost, and production FER advantage simultaneously. The strongest practical control in this pass is classical LMMSE-assisted ordered-statistics list generation followed by exact channel scoring. Its favorable small-system results must not be represented as invention of a new decoder. Modern soft-output OSD/GRAND/GCD work already addresses probability outside the list [R6]. The potential remaining contribution is a useful, affordable fidelity controller for the coupled SP-AFDM likelihood and its channel-refit statistics. The conditional theory below defines such a controller, but the present implementable worst-case bounds are often too conservative.

This conclusion follows an additional synthesis cycle rather than stopping at the previous candidate's limitations. First, exact code-induced likelihood simplification was investigated. Its large reductions on the tiny codes nearly disappear for the actual frozen code. Second, bounded-list refinement was implemented, tested at a second noise level on separate seeds, and checked against exact inference and diagonal-covariance controls. It has real diagnostic value, but neither its basic architecture nor lower total cost is established as a new contribution.

The main mathematical assets are: an exact characterization of code-induced quadratic equivalences; a posterior-mass/covariance decomposition; a score-oscillation transfer bound for outside-list odds; and a list-Gram-conditioned perturbation bound for the regularized channel update. These are proved under explicit conditions. They are not claims that every constituent identity is historically new.

2. What the v2 handoff actually supplies

2.1 Provenance and executable boundaries

All **44 manifest-listed files** match their supplied SHA-256 hashes and sizes. The added checkpoint matches `052ecfab8eff7780c49038d9bb4f16f92272ee22322fa05b6f65029cc3112a91`. Using the unmodified coding module, the frozen code was rebuilt with $(n, k) = (2048, 1024)$ and generator hash `bfd4247263adedc9a095b87603c2a7a4c2f2e8cb83f7e78bad721c5b06eaa6af`, exactly matching the fixture.

The smoke script was actually attempted with a namespace exposing every shipped module directory and the explicit supplied checkpoint. It stops at:

```
ModuleNotFoundError: No module named
'afdm_sp_joint_unrolled.budgeted_support_eval'
```

No replacement for this absent implementation was invented. The README itself says that the full repository is required. The fixture declares `a0`, `direct40`, support repair disabled, 10 outer iterations,

20 BP iterations, and LDPC maximum 50. This declaration is not sufficient to establish equivalence with the formal `TimingReceiver` branch that explicitly installs `CodeFeedback` and an outer-BP schedule. Because the actual `run_frame` is absent, equivalence remains unverified. The fixture's five failed frames out of eight are source-reported, not rerun results.

Role	Supplied assets	Treatment
Physical/system reference	`supplement_g1`, CPP/DAFT definitions, `e1_ldpc`	Physical function reused verbatim; independent dense check; code rebuilt
Frozen receiver	Receiver adapter, feedback modules, checkpoint, smoke fixture	Preserved; smoke attempted; incomplete runtime explicitly recorded
Historical exploration	Alternative estimators, support transactions, July/September reports	Evidence and comparator definitions, not commands to preserve an old narrative

The latest tail reports remain prior project evidence. Their 32-frame TAIL-versus-expanded-ridge comparison did not establish promotion. Their original arrays are not in this handoff, so those counts are not mixed with the new observations. The earlier likelihood-compression report is similarly an inherited analysis, not a substitute for this run's evidence.

2.2 Independent physical check

The handoff uses positive chirps and positive Doppler exponentials. With the negative-exponent unitary DFT,

$$F_{mn} = N^{-1/2} e^{-j2\pi mn/N}, \qquad A = \Lambda_{c_2}^{+} F \Lambda_{c_1}^{+}, \qquad \Lambda_c^{+} = \mathrm{diag}(e^{j2\pi c n^2}),$$

its path is $U(l,\nu) = AC_l\Delta_\nu^{+}\Pi_l A^H$. For a negative delayed index $t-l$, the prefix multiplier is $e^{j2\pi c_1[N^2+2N(t-l)]}$. These choices are sign-equivalent to the inherited manuscript after consistently reversing both chirp and Doppler parameters.

The original `physical_channel_matrix` function is executed as an unaltered AST function; unrelated unavailable imports are not emulated. A separately written construction builds the time-domain delayed matrix column by column, applies the prefix on wrapped samples, and conjugates by an explicit DAFT matrix. Nine cases include $N = 8, 12, 9$, fractional and integer Dopplers, and a nontrivial prefix parameter. The maximum entrywise disagreement is 3.99×10^{-15}.

3. Model and information contracts

Let $b \in \mathbb{F}_2^k$ be an information message, $c = Gb \pmod 2$ its binary codeword, and $d(c) \in \mathbb{C}^N$ its QPSK vector, with $n = 2N$. The system is

$$x = p + d(c), \qquad y = H(\theta)x + w, \qquad w \sim \mathcal{CN}(0, \sigma^2 I_N). \tag{1}$$

The deterministic pilot and data already include their power scaling. In the experiments, $|p_i|^2 = \rho$ and $|d_i|^2 = P_d = 1 - \rho$. A particular block's pilot-data cross term need not vanish. Neither the full channel nor a receiver approximation is rescaled after truncation.

The principal physical assumptions are integer sample delays, full fractional Doppler, fixed path parameters within the block, sufficient CPP, ideal synchronization, and proper sampled white noise. Fractional delays, arbitrary pulse shaping, hardware impairments, and channel drift within a frame require extensions. Each path operator is unitary, but the multipath sum generally is not.

The complete objective remains information-frame error subject to full receiver cost. The following objects are kept distinct: physical H; an integer representation of H; a frozen receiver operator $\bar{H}$; a working code posterior; its list approximation; a frozen-distribution channel refit; and the final iterative receiver risk.

The finite identity

$$U(l,\nu)=\sum_{q=0}^{N-1} c_q(\nu)U(l,q), \qquad c_q(\nu)=\frac{1}{N}\sum_{t=0}^{N-1} e^{j2\pi(\nu-q)t/N} \tag{2}$$

means that integer representation is not inherently approximate. Restricting its support is approximate. Contributions from different physical paths sharing an integer coordinate must first be added coherently. Effective-shift collisions, in contrast, concern the map from a coordinate to its occupied DAFT diagonal and do not imply identical physical operators.

For a fixed $\bar{H}$, set

$$R=\bar{H}^H\bar{H}, \qquad z=\bar{H}^H(y-\bar{H}p).$$

The exact plug-in codeword score is

$$s(c)=\frac{2\mathfrak{R}[d(c)^H z]-d(c)^H R d(c)}{\sigma^2}+\log\pi(c), \qquad P(c)=\frac{e^{s(c)}}{\sum_{c'\in\mathcal{C}} e^{s(c')}}. \tag{3}$$

When $\bar{H}=H$ is known and the prior is correct, P is the true message posterior. When $\bar{H}$ is estimated from the same y, it is a frozen plug-in distribution. Algebraic comparisons after freezing remain valid, but they do not integrate channel uncertainty or automatically yield true-posterior risk guarantees.

The previous common-Doppler invariance $H_\delta=V_\delta H_0$ and $H_\delta^H H_\delta=H_0^H H_0$ remains valid. Its favorable compression is code-independent and can be explained by ordinary derotation. The code audit below does not refute that result.

4. First synthesis remedy: exact code-induced likelihood reduction

4.1 Character representation

Write $s_i=(-1)^{c_i}$ for binary signs, and let g_i be row i of G. For a coordinate subset S, define

$$\chi_S(b)=\prod_{i\in S} s_i=(-1)^{(\bigoplus_{i\in S} g_i)\cdot b}. \tag{4}$$

Proposition 1 (exact equivalence classes). Two products χ_S and χ_T agree for every valid message if and only if

$$\bigoplus_{i\in S} g_i = \bigoplus_{i\in T} g_i,$$

or equivalently the incidence vector of $S\triangle T$ belongs to the dual code $\mathcal{C}^{\perp}$.

Proof. Their product is the character indexed by the XOR of the two signatures. The zero signature is identically one. A nonzero binary linear form takes each value equally often over uniformly distributed information messages and therefore is not identically zero. This proves both directions. The dual statement is the definition of a parity relation holding on every codeword. □

For QPSK, set $a = \sqrt{P_d/2}$, $d = a(s_I + js_Q)$, and

$$B_R = \begin{bmatrix} \Re R & -\Im R \\ \Im R & \Re R \end{bmatrix}.$$

Apart from the code prior, (3) is a real polynomial of degree at most two in the signs. Its linear coefficients are $2a[\Re z; \Im z]/\sigma^2$, pair coefficients are $-2a^2(B_R)_{ij}/\sigma^2$, and the diagonal is constant. Coefficients with equal XOR signatures can be added exactly. The independent verifier reconstructs every small-system score this way and checks it against direct complex Euclidean distances.

Corollary 1 (dual-distance obstruction). If the dual minimum distance is at least five, all constant, singleton, and distinct-pair characters are distinct. Consequently, no nonzero polynomial built from these characters can vanish identically on the code. There is no additional universal exact deletion or merging of degree-two terms beyond constant/diagonal identities.

Proof. The symmetric difference of two different sets of size at most two has weight between one and four, contradicting the dual-distance condition. Distinct binary characters are orthogonal under the uniform message distribution, so their coefficients cannot cancel identically. □

This is a standard coding-character argument applied to the receiver, not an invention of Fourier duality. It excludes a particular exact simplification, not approximate pruning, posterior concentration, a different physical basis, or all fast decoders.

4.2 Why tiny-code evidence does not extrapolate

The exact audit considers the generic real feature dictionary with $1 + n + \binom{n}{2}$ terms. It does not claim that every term has a nonzero coefficient in a particular AFDM channel.

Code length n	Dimension k	Check size setting	Generic terms	Exact classes	Redundant fraction
16	6	4	137	40	70.8029%
24	8	4	301	97	67.7741%
96	50	6	4,657	4,657	0%
Frozen 2,048	1,024	6 with short edge rows	2,098,177	2,092,033	0.292826%

The frozen construction has 1,023 weight-6 checks and three weight-2 checks, since 2,048 is not divisible by six. It has 2,045 distinct coordinate characters. Thus it would be incorrect to assume dual distance at least five for this particular code; the audit handles its actual exceptions.

A 64-bit linear projection is used only to bucket XOR signatures. Every colliding bucket is resolved using the full generator rows. Exact signature equality always implies equal projected signatures, so the audit has no false-negative equality risk. It is not a probabilistic declaration that hash collisions never occur. No spurious collision bucket occurred here. The frozen-code pass takes about 0.82 seconds in the recorded environment.

Decision: exact code quotienting is a valid reference optimization but is not selected as the main practical innovation. The spectacular tiny-code reduction does not establish a similar benefit for the production code. Character counts are not a FLOP or wall-time reduction claim.

5. The overlooked consumer: channel-estimation moments

For active integer operators $U_1, \dots, U_S$, define

$$H(g) = \sum_{i=1}^{S} g_i U_i, \qquad \Phi(d) = [U_1(p+d), \dots, U_S(p+d)].$$

A fixed working distribution q gives

$$J_q(g) = \mathbb{E}_q \|y - \Phi(d)g\|^2 + \lambda \|g - g_0\|^2, \qquad \lambda > 0. \tag{5}$$

With $\mu_q = \mathbb{E}_q d$, $V_q = \mathrm{Cov}_q(d)$,

$$K_q = \mathbb{E}_q[\Phi(d)^H \Phi(d)] = \Phi(\mu_q)^H \Phi(\mu_q) + \Gamma_q, \qquad (\Gamma_q)_{ij} = \mathrm{tr}(U_i^H U_j V_q), \tag{6}$$

and $b_q = \mathbb{E}_q[\Phi(d)^H y]$, the minimizer is

$$g_q = (K_q + \lambda I)^{-1}(b_q + \lambda g_0). \tag{7}$$

These are the inherited frozen-distribution identities. They require no Gaussian assumption about q. With diagonal V_q, unit-modulus monomial operators, and distinct effective shifts, $\Gamma_q = \mathrm{tr}(V_q) I$. The coefficient error depends on the whole normal-equation pair (K_q, b_q), not on Γ_q alone when the mean changes.

Proposition 2 (MAP is insufficient for CE moments). Exact agreement of the maximizing codeword gives no general guarantee that μ, V, Γ, or (7) are preserved.

Proof by construction. Keep only a maximizing codeword and normalize its mass to one. The maximizer is unchanged, but the resulting covariance and Γ are zero. Any nondegenerate original posterior has nonzero covariance; its projection through Γ is nonzero whenever one retained operator maps a varying symbol vector to a varying response. □

An executed witness has $N = 12$, the three-path regime, seed 29, $\rho = 0.5$, noise variance 0.3, and frame 13. Classical OSD with a singleton list returns the correct global MAP word. Nevertheless, that word has posterior probability only 0.583518654. The list discards 0.416481346 mass, sets Γ to zero instead of a matrix with Frobenius norm 0.862689366, and changes the regularized coefficient vector by 0.142564200. This is a selected counterexample, not a population-average effect.

A normalized in-list effective sample size is also insufficient: a singleton always has ESS one, whether it contains almost all posterior mass or very little. Small ESS does not alone prove failure either; a genuinely concentrated complete posterior can have ESS near one.

6. Second synthesis remedy: full-score short lists with moment control

6.1 Exact list identities

For a nonempty list $\mathcal{L} \subseteq \mathcal{C}$, let

$$\epsilon = P(\mathcal{C} \setminus \mathcal{L}), \qquad P_L = P(\cdot \mid \mathcal{L}), \qquad P_T = P(\cdot \mid \mathcal{L}^c).$$

Proposition 3 (mass, risk, and covariance). For $0 \le \epsilon < 1$,

$$\mathrm{TV}(P, P_L) = \epsilon, \qquad D_{\mathrm{KL}}(P_L \| P) = -\log(1-\epsilon). \tag{8}$$

The conditional excess message risk of the full-score best-in-list decision over full MAP is at most ϵ. The covariance obeys

$$V = (1-\epsilon)V_L + \epsilon V_T + \epsilon(1-\epsilon)(\mu_L - \mu_T)(\mu_L - \mu_T)^H. \tag{9}$$

Consequently,

$$\Gamma = (1-\epsilon)\Gamma_L + \epsilon\Gamma_T + \epsilon(1-\epsilon)\mathcal{G}((\mu_L - \mu_T)(\mu_L - \mu_T)^H), \tag{10}$$

where $\mathcal{G}(V)_{ij} = \mathrm{tr}(U_i^H U_j V)$ is a positive linear map.

Proof. On the list, $P = (1-\epsilon)P_L$; off the list, $P_L = 0$. Sum the absolute probability differences to obtain (8), and substitute the constant density ratio for KL. If full MAP lies in the list, the decisions agree. Otherwise its probability is at most ϵ, so its advantage over the best retained word is at most ϵ. Equation (9) is obtained by expanding second moments of the two-component mixture and subtracting the mixture mean outer product. Positivity of $\mathcal{G}$ follows because $g^H \mathcal{G}(V) g = \mathrm{tr}(H(g) V H(g)^H) \ge 0$. □

In particular, $\Gamma \succeq (1-\epsilon)\Gamma_L$, but $\Gamma \succeq \Gamma_L$ need not hold. Enlarging a list need not change covariance monotonically. The full-list endpoint has $\epsilon = 0$. Probabilities close to zero are calculated by summing the complement, not by subtracting an almost-unit in-list sum.

6.2 Projected moments without an $N \times N$ covariance

For weighted words d_r and mean μ_L, form

$$B_r = [U_1(d_r - \mu_L), \ldots, U_S(d_r - \mu_L)].$$

Then

$$\boxed{\Gamma_L = \sum_{r \in \mathcal{L}} w_r B_r^H B_r.} \tag{11}$$

This centered implementation is positive semidefinite by construction. For monomial operators, a direct implementation costs $O(LNS^2)$ after obtaining the list and weights. It avoids storing the full covariance but does not make list acquisition free. Diagonal moments reduce the estimator to scalar ridge under the noncollision assumptions and remain an essential simple control.

6.3 Transfer of outside-list probability between likelihood models

Let a proxy score t define $Q \propto e^t$ on the same codebook. Define $e = s - t$, $\Omega = \max e - \min e$, and $\delta = Q(\mathcal{L}^c)$.

Proposition 4 (outside-list odds transport).

$$\boxed{\epsilon \le \frac{e^{\Omega}\delta}{1-\delta+e^{\Omega}\delta}.} \tag{12}$$

The lower bound is obtained by replacing Ω with $-\Omega$.

Proof. For an interior list mass,

$$\frac{P(\mathcal{L}^c)}{P(\mathcal{L})} = \frac{Q(\mathcal{L}^c)}{Q(\mathcal{L})}\frac{\mathbb{E}_Q[e^e \mid \mathcal{L}^c]}{\mathbb{E}_Q[e^e \mid \mathcal{L}]}.$$

The expectation ratio lies between $e^{-\Omega}$ and e^{Ω}. Rearrange the upper odds inequality; the endpoint cases follow directly. □

For Gram pruning with exact z and $E = R - R_B$, a computable sufficient range is

$$\Omega \le \overline{\Omega}_B = \frac{4P_d}{\sigma^2}\sum_{i<j}|E_{ij}|, \tag{13}$$

after discarding codeword-constant diagonal contributions. A code-character grouping can tighten the range. A spectral alternative is $NP_d[\lambda_{\max}(E) - \lambda_{\min}(E)]/\sigma^2$. These remain worst-case bounds. If the SP linear statistic is also approximated, its score variation must be added; it cannot be silently ignored.

A soft-output estimate of δ is not automatically an upper bound. The guarantees require a valid bound under the proxy model, then (12) under the actual frozen score. Modern SO-OSD/GCD/GRAND formulas use explicit modeling assumptions [R6]; the implemented experiment does not claim to reproduce those published algorithms. It uses exact short-code proxy normalization as a diagnostic. Applying a memoryless confidence estimate directly to a coupled Gram likelihood is not justified by its good calibration in a different model.

6.4 A refit certificate using the list's own information matrix

Assume unitary U_i and a known radius $\|p+d\| \le R_x$. For QPSK, the inexpensive bound $R_x = \|p\| + \sqrt{NP_d}$ suffices. Let g_L minimize (5) under P_L, and let K_L be its unregularized matrix in (6).

Proposition 5 (list-Gram-conditioned CE bound). With $m_L = \lambda_{\min}(K_L) \ge 0$,

$$\boxed{\|g_P - g_L\| \le \frac{\epsilon\left[\sqrt{S}R_x(\|y\| + \sqrt{S}R_x\|g_L\|) + \lambda\|g_L - g_0\|\right]}{\lambda + (1-\epsilon)m_L}.} \tag{14}$$

A valid upper bound on ϵ may replace it everywhere on the right.

Proof. By mixture linearity, $K_P = (1-\epsilon)K_L + \epsilon K_T$ and $b_P = (1-\epsilon)b_L + \epsilon b_T$. Subtract the list normal equation from the complete one to obtain

$$(K_P + \lambda I)(g_P - g_L) = \epsilon\{b_T - K_T g_L - \lambda(g_L - g_0)\}.$$

Since $K_T \succeq 0$, the minimum eigenvalue on the left is at least $\lambda + (1-\epsilon)m_L$. Also

$$\|\Phi(d)\| \le \sqrt{S}\|p + d\| \le \sqrt{S}R_x,$$

so $\|\Phi(d)^H[y - \Phi(d)g_L]\| \le \sqrt{S}R_x(\|y\| + \sqrt{S}R_x\|g_L\|)$. Average over P_T, apply the triangle inequality, and divide by the eigenvalue lower bound. The right-hand side is nondecreasing in $\epsilon \in [0,1]$. □

Compared with a generic $1/\lambda$ perturbation bound, (14) exploits information already available in the list Gram matrix. It can be much tighter. Its guarantee concerns the same support, data model, noise units, proximal center, and regularization, with the working distribution frozen. It is not an EM-ascent statement for approximate feedback, nor an FER-descent result. When $\lambda = 0$, additional nonsingularity conditions are needed; the stated proposition deliberately excludes that case.

For an observation-dependent $\bar{H}(y)$, equations (8)-(14) compare finite distributions and refits after freezing y. They do not require independent training. Their reference is then the full plug-in distribution. If $P_\star$ is the actual posterior, its discrepancy from $P_{\bar{H}}$ is a separate uncontrolled term. A bound on list error cannot remove it.

7. Reference algorithm and what is deployable

The practical candidate uses a classical reliability-based OSD proposal, not a full codebook search. The implemented OSD selects a most-reliable independent generator-row basis, performs GF(2) reprocessing, orders test patterns of weight at most two by reliability cost, and retains at most L words. The full AFDM operator then scores those words by (3). No true bits or gains are accepted by the proposal function.

A theoretical guarded variant operates as follows:

1. Freeze the current operator, receiver messages, active integer support, regularization, and a baseline fallback state.
2. Produce a bounded candidate list by a declared classical proposal. Compute full-channel scores on the list and their normalized weights.
3. Compute μ_L, Γ_L by (11), K_L, and g_L.
4. Acquire an independently justified upper bound on outside-list mass under an explicit proxy score, transfer it using (12), and evaluate (14).
5. Accept a tolerance-certified refit, expand the list within a fixed budget, or retain the declared fallback. A complete-code fallback is permitted only in the small reference. No certificate is reported when a required probability bound is unavailable.

The code implements the proposal, rescoring, moments, and certificate. The short-code study supplies exact probability masses for diagnostic verification. **A scalable certified outside-list estimator for the full coupled likelihood is not supplied.** Treating the exact short-code mass as a free deployable input would be oracle leakage in the complexity claim.

For nested lists, retained probability mass is nondecreasing and complete-code fallback terminates, but no monotonic FER improvement follows. If the fallback is an existing receiver state rather than full enumeration, only accepted steps receive the conditional refit guarantee. Repeated decoder feedback must retain the project's extrinsic/replacement conventions; list probabilities cannot simply be accumulated as new independent evidence.

Straightforward costs are: dense LMMSE proposal preparation $O(N^3)$ or an explicitly structured solver; observation processing $O(N^2)$; GF(2) OSD reprocessing up to $O(nk^2)$; up-to-order-two pattern generation $O(k^2)$ before truncation in this implementation; exact list scoring $O(LN^2)$ for dense $\bar{H}$, or the appropriate operator cost; projected moments $O(LNS^2)$ for monomial atoms; and $O(S^3)$ coefficient solve/eigenvalue computation. Shared work must be charged once, but never omitted entirely. The provided OSD pattern generator is not optimized for $k = 1024$.

8. Executed experiments and skeptical checks

8.1 Scope and pairing

The main grid uses $N = 8, 12$, the unmodified tiny Gallager constructions $(n, k) = (16, 6), (24, 8)$ with check-size setting four, three path regimes, $\rho \in \{0.25, 0.5\}$, noise variance 0.3, two physical phase seeds, and 16 observations per cell. It has 384 configuration/observation instances. A skeptical pass uses separate seeds, variance 0.6, and eight observations per cell, giving 192 more instances. Neither grid is a production-length LDPC test. Reusing exogenous messages/noise across matched power/regime comparisons prevents counting every configuration as independent fading evidence.

Propagation always uses the full fractional operator. The regimes are two paths with common fractional offset, two distinct fractional offsets, and three paths with distinct offsets. Gains are normalized once at generation and shared by every arm. The primary tests give every detector the same actual H to isolate posterior approximation. Channel-refit diagnostics use the same three fixed integer atoms with distinct effective shifts and $\lambda = 0.2$; this support is intentionally not the full physical model.

Why refit after supplying an oracle channel? This is a controlled interface experiment: it asks whether different posterior approximations feed the same subsequent integer refit consistently. It is not a recommendation to replace known CSI with an estimated channel. The resulting post-refit FER is saved but is not used as a new receiver-performance claim.

8.2 Terminal decision results

Arm	Main failures / 384	Main disagreements with full MAP	Higher-noise failures / 192	Higher-noise disagreements
Full likelihood	32	0	46	0
Magnitude-pruned Gram, $B = N$ pairs	46	25	58	23
OSD list 1 + full scoring	71	60	69	53
OSD list 4 + full scoring	44	20	48	13
OSD list 16 + full scoring	32	0	46	0
Best four proxy-score words + full scoring	32	0	46	0

The last row requires exact small-code proxy enumeration and is an oracle proposal-quality control, not

a low-cost decoder. OSD uses the same underlying channel but an implementable bounded proposal. Equal list sizes mean equal final full-score counts, not equal total cost or equal proposal quality. The fixed-Gram comparison uses a different resource model and is not called equal-cost.

Zero disagreements on these grids are descriptive, not proof of population equivalence. List 16 is a classical strong control that should be retained in a future comparison; it is not a newly invented algorithm.

8.3 Posterior mass and moment diagnostics

OSD budget	Main mean omitted mass	Higher-noise mean omitted mass	Main mean coefficient change	Higher-noise mean coefficient change
1	0.182216	0.342968	0.0762481	0.152020
4	0.0535591	0.103284	0.0228497	0.0430246
16	0.00121983	0.0122592	0.000523647	0.00475184

The reference for coefficient change is the complete-posterior version of the same regularized refit, not the true channel. At higher noise, list 16 still matches every full MAP decision but has maximum omitted mass 0.325660. This isolates the difference between selecting a winner and retaining uncertainty.

An exact full-posterior-mean plus diagonal-covariance control changes coefficients by 0.00335207 on average in the main grid and 0.0117639 at higher noise. This confirms that off-diagonal moments can matter, but it does not prove they are worth acquiring in the production receiver. Relative Γ errors can become very large when the reference covariance is nearly zero; absolute moment errors and coefficient changes are therefore retained alongside relative metrics.

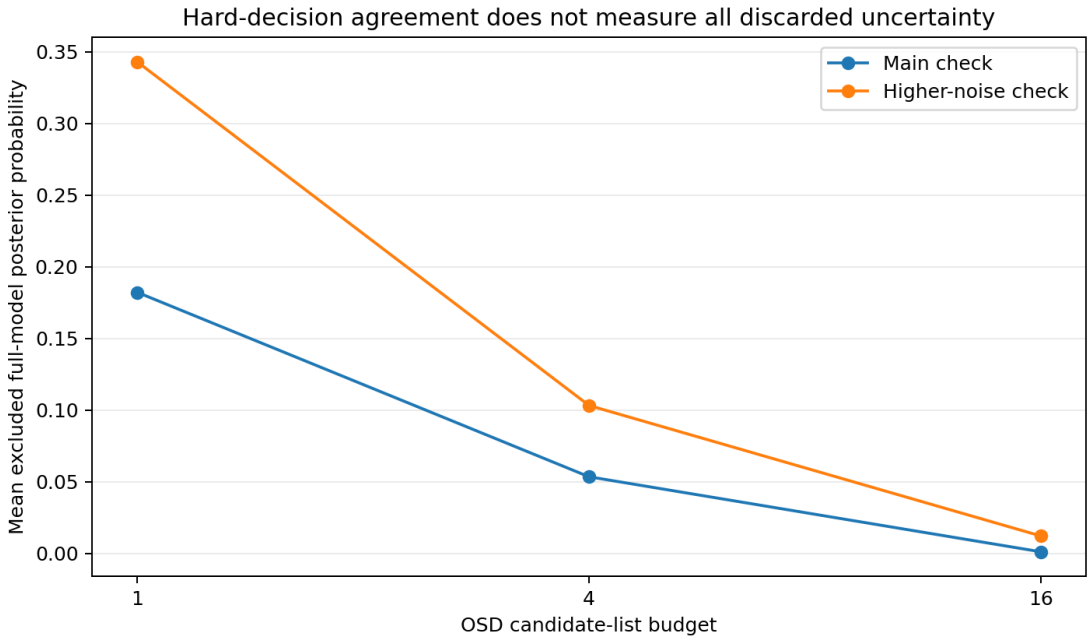


Figure 1: Descriptive missing-mass comparison. These are true full-model masses available only in the small exact reference.

8.4 Proxy confidence is not transferable without a likelihood correction

A four-word list in the $N = 8$ three-path case, seed 11, $\rho = 0.5$, frame 11, has proxy outside mass only 0.006848. Under the full likelihood its outside mass is 0.286605. Another $N = 12$ example changes from 0.000605 to 0.123811. These are explicit counterexamples to using the proxy's confidence unchanged.

The transfer bound (12) remains valid, but the global range bound can be large. It often sacrifices useful coverage to protect against adverse codewords. This is not a bug repaired by taking more Monte Carlo samples.

For an absolute coefficient tolerance of 0.01:

OSD-16 outcome	Main / 384	Higher-noise / 192
Actual coefficient change within tolerance	379	174
Certified by (14) using exact omitted mass	355	109
Certified after worst-case proxy-mass transfer	122	39

The independent presence of good list moments and a poor inexpensive certificate identifies the method gap. Exact omitted mass is not a practical oracle to assume away. Improved soft-output models must retain their own assumptions and be tested under the coupled likelihood.

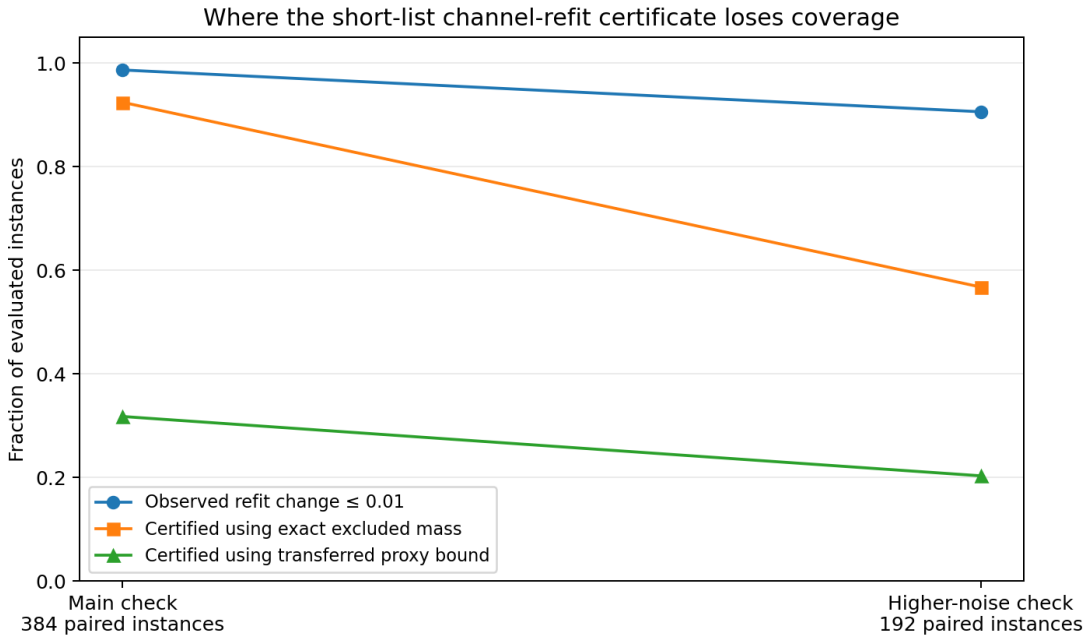


Figure 2: Certificate coverage for a fixed 0.01 coefficient tolerance. The exact-mass curve is an oracle diagnostic, not deployable evidence.

8.5 Observation-dependent channel check

A separate 24-frame diagnostic obtains $\bar{H}(y)$ from pilot-only ridge fitting on the fixed integer support. The initializer reads only y, p, U, σ^2. Both full and list posteriors use that same estimate. All 24 refit perturbation inequalities pass; full-plugin and list decoding each fail on 20 frames. This is a poor estimator used to test pointwise applicability, not a competitive baseline. It confirms that algebraic validity survives same-observation estimation while channel-estimation accuracy remains a separate bottleneck.

8.6 Timing and independent verification

The dedicated benchmark excludes truth scoring, complete-code diagnostics, and saved-array audit work. It includes the specified online proposal, full scoring, moments, and CE operations. Common channel preparation and full-code response-template preparation are separately recorded.

Kernel, median of 31 repeats	$N = 8$	$N = 12$
Full posterior + projected moments + CE	136.666 microseconds	329.266 microseconds
Full posterior + diagonal moments + CE	45.438 microseconds	66.660 microseconds
OSD-4 + scoring + projected moments + CE	337.999 microseconds	373.873 microseconds
OSD-16 + scoring + projected moments + CE	362.676 microseconds	456.317 microseconds

Full enumeration is feasible only because the codebooks are tiny. These numbers neither prove production enumeration is practical nor prove list methods cannot win there. They do refute claiming a demonstrated speed advantage for this implementation. The faster diagonal control is especially relevant to the project's existing scalar-Γ interface.

The independent verifier imports none of the simulator, list-refinement, or common utility modules. It recomputes direct Euclidean codeword likelihoods, binary-character scores, centered response Grams, and rectangular augmented least-squares solutions. **12,096 assertions pass**, with maximum absolute discrepancy 1.42×10^{-13}. A separate same-environment rerun reproduces **16,392 arrays in 24 primary raw files exactly**. These assertion/array counts are engineering checks, not independent scientific replications. Python 3.13.5 and NumPy 2.3.5 are recorded; timings are machine-specific.

9. Related-work and novelty boundary

Closest work	Relevant overlap	Consequence for this study
Zheng et al., SP-AFDM [R2]	Pilot placement; iterative channel estimation and detection	SP plus iteration is baseline methodology
Huang et al., SP-OTFS [R3]	BEM representation and soft message-passing estimation/detection	Integer/BEM models and soft second moments are not new
AFDM MF-MP, DSP 2024 [R4]	Matched-filter-domain sparsity for detection	The previous Gram-domain transformation is already occupied
Joint Sparse Graph, MIMO-AFDM [R5]	Joint detection/LDPC graph and weak-edge simplification	Coded graph sparsification is not a new architecture
Feng, Duffy, and Medard [R6]	Outside-list probability and improved soft output for OSD/GCD/GRAND/SCL	List-confidence correction itself is not novel; its assumptions must be respected

Closest work	Relevant overlap	Consequence for this study
Luo et al., data-aided AFDM-ISAC SBL [R7]	SP, fractional delay/Doppler, reliable data feedback, grid evolution and GAMP	Broad combination of SP, fractional channels and data-aided sparse estimation is occupied
Mao et al., AFDM chirp/receiver optimization [R8]	Structured sparsification under a complexity constraint	Generic complexity-aware AFDM pruning is not enough

The new derivations were developed independently, but they combine established probability conditioning, coding-character algebra, and strongly convex least-squares perturbation tools. They are not accompanied by an exhaustive originality certificate. The most defensible potential distinction is **controlling posterior error in the actual code-to-estimator sufficient statistics under the coupled SP-AFDM likelihood**, rather than preserving only a hard decision or calibrating a memoryless list decoder. The current bound's conservatism and missing practical probability acquisition mean that this distinction has not yet become a demonstrated superior method.

This search inspected public primary papers and publisher records available through the research date. Abstract-only publisher access is sufficient to identify overlap but not to claim a faithful reproduction of every published receiver. No runtime or FER number from a paper is silently compared with the different small system used here.

10. Claim status, remaining experiment, and stopping decision

Proved: finite-model likelihood/character identities; dual-distance obstruction; list TV/risk/covariance relations; missing-mass odds transport; projected covariance positivity; and the regularized refit bound.

Executed and supported only in the declared scope: v2 hash/code reconstruction; nine dense physical checks; tiny-code versus frozen-code equivalence counts; terminal and moment comparisons on 576 paired oracle-model instances; a 24-frame estimated-model check; the timing comparison; independent saved-array verification; and deterministic primary replay.

Not established: a new production I1 improvement, affordable trustworthy full-code correlations, a tight scalable missing-mass certificate for the coupled likelihood, an end-to-end low-cost advantage, or an independently novel Transactions contribution. A correct support or moment identity is not a substitute for those results.

The remaining driver `src/confirmation_driver.py` consumes actual frozen-receiver terminal snapshots and compares the unchanged terminal output with a classical bounded-list full-score control. It is executable, documents its exact input contract, and is not run because the required snapshots are absent. It is a terminal comparison, not a hidden substitute for a complete feedback integration. The separate small-model expansion command and production prerequisites are recorded in `docs/confirmation_protocol.md`.

The selected line remains posterior-fidelity-controlled code-aided CE. The exact code quotient is retained as a reference simplification, not the main method. Unqualified MAP-only certification is insufficient for joint reception. OSD list refinement is retained as a strong classical comparator. Off-grid estimation, structured MMSE, and decoder refresh remain necessary controls, not disproved approaches. Rich correlations remain conditional on useful acquisition cost. No learning module is warranted by this evidence.

The present stopping point is scientific: the central mechanism, counterexample, and bound are checked independently; two different remedies have been examined; the strongest simple explanations and direct prior art have been identified. More short-code samples would not fix missing originality, an overconservative certificate, or an absent full receiver interface.

References and source record

[R1] A. Bemani, N. Ksairi, and M. Kountouris, "Affine Frequency Division Multiplexing for Next Generation Wireless Communications," IEEE TWC, 2023. Author manuscript arXiv:2204.12798. Background inherited from the supplied theoretical chapter.

[R2] K. Zheng, M. Wen, T. Mao, L. Xiao, and Z. Wang, "Channel Estimation for AFDM With Superimposed Pilots," arXiv:2404.10232; IEEE TVT 74(2), 3389–3394, 2025. Public record: https://arxiv.org/abs/2404.10232.

[R3] F. Huang, Q. Guo, Y. Zhang, and Y. Zakharov, "Message Passing-Based Joint Channel Estimation and Signal Detection for OTFS with Superimposed Pilots," arXiv:2309.08177v2. Full text inspected: https://arxiv.org/html/2309.08177v2.

[R4] "AFDM signal detection based on message passing scheme," Digital Signal Processing 153, 104633, October 2024. DOI 10.1016/j.dsp.2024.104633. Publisher abstract inspected: https://www.sciencedirect.com/science/article/pii/S1051200424002586.

[R5] "Joint Sparse Graph for Enhanced MIMO-AFDM Receiver Design," arXiv:2503.19143v1. Full text inspected: https://arxiv.org/html/2503.19143v1.

[R6] J. Feng, K. R. Duffy, and M. Medard, "Leveraging Code Structure to Improve Soft Output for GRAND, GCD, OSD, and SCL," arXiv:2503.16677v1, 20 March 2025. In particular Sections II–III distinguish normalized in-list probabilities from estimates including unvisited codewords, under their explicit channel/code modeling assumptions. https://arxiv.org/html/2503.16677v1.

[R7] Y. Luo et al., "Data-aided Channel Estimation and Sensing With Sparse Bayesian Learning for AFDM-ISAC System," arXiv:2607.18881v1, 21 July 2026. https://arxiv.org/html/2607.18881v1.

[R8] R. Mao et al., "Joint Chirp Parameter Selection and Low-Complexity MMSE Receiver Design for AFDM Systems," arXiv:2607.20227, 22 July 2026. https://arxiv.org/abs/2607.20227.

[P1] Supplied `spafdm_receiver_handoff_20260917-v2.zip`: README, MANIFEST, RUNTIME_CLOSURE, physical/coding modules, checkpoint, smoke fixture, state/derivation notes, and September tail reports. Hash verification in `results/input_audit.json`; original bytes retained under `inputs/`.

[P2] Supplied "A Consistent Theory of Coded SP-AFDM Reception," 19 September 2026; conditional model, fixed-q CE, and evidence boundaries retained. Its earlier theorem numbers are not reused for the new propositions.

[P3] Previous "Likelihood-Preserving Compression for Coded SP-AFDM" report from this conversation. Its hard-decision certificate and cost limitations motivated the skeptical pass; no old raw FER values are relabeled as newly executed data.